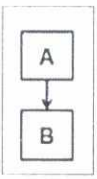
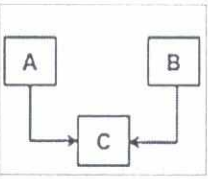
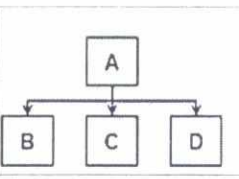
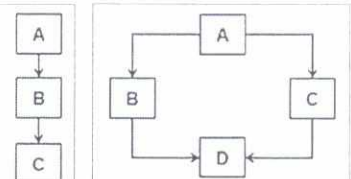


Roll No:

Thapar Institute of Engineering & Technology, Patiala			
Mid Semester Examination			
UTA018: Object Oriented Programming			
B. Tech. (4 th Sem)	Time: 2 hrs	Marks: 30	Date: 03.03.25
Faculty: Dr. Deep Mann, Dr. Seemu Sharma, Dr. Saurab Arora			
Instructions: Assume any missing data. Attempt all the questions in sequence. Start new question from fresh page. Attempt all the parts of question together.			
Q1	Read the code carefully and write the output of the code after filling the blanks. No need of writing the whole program just write the line number for fill ups and output.		3+3+4
Q1(A)	<pre> 1. #include <iostream> 2. using namespace std; 3. class myclass{ 4. int a, b; 5. public: 6. friend int sum(myclass x); 7. void set_ab(int i, int j); 8. }; 9. void myclass::set_ab(int i, int j){ 10. a = i; 11. b = j; 12. } </pre>	<pre> 13. int sum(myclass x){ 14. return x.a + x.b; 15. } 16. int main(){ 17. myclass n,n1,n2; 18. n.set_ab(3, 4); 19. n1.set_ab(6,10); 20. n2.set_ab(50,70); 21. cout <<"First Sum"<< _____ <<endl; 22. cout <<"Second Sum"<< _____ <<endl; 23. cout <<"Third Sum"<< _____ <<endl; 24. return 0; 25. } </pre>	(CO1) (BL2)
Q(B)	<pre> 1. #include<iostream> 2. using namespace std; 3. class Test 4. { 5. public : 6. int x; 7. void show_data(); 8. }; 9. void Test _____ 10. { 11. cout<<"\n x="<<x; 12. } 13. int main() 14. { Test t; 15. int Test :: *ptr= _____; 16. t.*ptr=20; 17. t.show_data(); 18. _____ = new Test; 19. tp->*ptr=80; 20. _____ show_data(); 21. return 0;} </pre>	Q(C)	<pre> 1. #include <iostream> 2. using namespace std; 3. class cl 4. { public: 5. cl(int i) { val=i; } 6. int val;int double_val() 7. { return val+val; } 8. }; 9. int main() 10. {int cl::*data; 11. int (cl::*func)(); 12. cl ob1(1), ob2(2); 13. data = _____; 14. func = _____; 15. cout << "Here are values: "; 16. cout << ob1.*data << " " << ob2.*data << "\n"; 17. cout << "Here they are doubled: "; 18. cout << _____ << " "; 19. cout << _____ << "\n"; 20. return 0;} </pre>

Q2	Write a C++ program that defines a class Student with the following members: (a) A default constructor that initializes the student's name to "Unknown" and age to 0. (b) A parameterized constructor that accepts the student's name and age as arguments. (c) A display function that prints the student's details. (d) A destructor to destroy the object. (e) In the main () function, create two objects—one using the default constructor and another using the parameterized constructor. Based on the above program, answer the following questions: (i) Explain the order of execution of constructors and destructors in this program. (ii) How can we modify the code so that obj3's destructor is never called automatically?	6+4 (CO1) (BL3,4)
Q3	Inheritance is a fundamental concept in object-oriented programming (OOP) that allows a class to inherit properties and behaviours (methods) from another class. Identify the different types of inheritance shown below. Write the syntax of each type and also explain the access specifiers used in inheritance.  <pre> graph TD A[A] --> B[B] </pre>  <pre> graph TD A[A] --> C[C] B[B] --> C[C] </pre>  <pre> graph TD A[A] --> B[B] A[A] --> C[C] A[A] --> D[D] </pre>  <pre> graph TD A[A] --> B[B] A[A] --> C[C] B[B] --> D[D] C[C] --> D[D] </pre>	2+5+3 (CO2) (BL1)